

Lab 3 — Architectural Justification

Smart Lab Equipment & Slot Reservation Portal

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Architecture Selection: *“We chose Layered Architecture for the Smart Lab Equipment & Slot Reservation Portal System.”*

Architectural Choice

We selected the Layered Architecture, organizing the system into a Presentation Layer (Portal UI Component), a Business Layer (Reservation Manager, Calibration Service, and Notification Service Components), and a Data Layer (Equipment & Reservation Database Component).

Two Scenario-Specific Reasons

1. Clean separation for a rule-driven workflow. The portal's core flow — view availability → reserve → verify calibration → generate token — is a well-defined sequential process. Layering isolates the student/technician-facing UI from the booking and calibration rules (FR-001–FR-005) and from persistent storage, so a rule change, such as adjusting the late-return hold threshold (FR-005), touches only the Business Layer's Reservation Manager Component and nothing else.

2. Supports a single, strict concurrency lock. NFR-001 requires concurrent slot-reservation requests to resolve within 200 ms under one strict database-level lock. Keeping the Reservation Manager and Database Components within the same layered application boundary (rather than splitting them into independently deployed microservices) avoids extra network hops and distributed-consistency overhead, making a single authoritative lock per equipment slot far easier to guarantee.

Security Advantage

Every request is routed through the Business Layer, so the Data Layer — equipment inventory, calibration records, reservation tokens, and late-return counts — is never exposed directly to the Presentation Layer or to student/technician clients. Reads and writes must pass through the Reservation Manager's and Calibration Service's validation logic, closing off direct database access and reducing the risk of tampering with calibration status or reservation tokens.

Performance Benefit

Because the layers operate within one tightly coupled application boundary rather than across independently deployed services, calls between the Reservation Manager and the Database Component avoid inter-service network latency. This keeps slot-lock and calibration-check operations comfortably within NFR-001's 200 ms target and helps sustain NFR-002's 99.5% uptime requirement, since there are fewer independently failing network boundaries than a microservices deployment would introduce.